

L03 – Balanced (Single-phase Equivalent) Power Flow

Foundations Lecture Series

Course Instructor

Microgrid 3-phase Security AI Project

Foundations Lecture 3

Keywords: balanced operation, phasor symmetry, single-line diagram, positive sequence, when balance lies

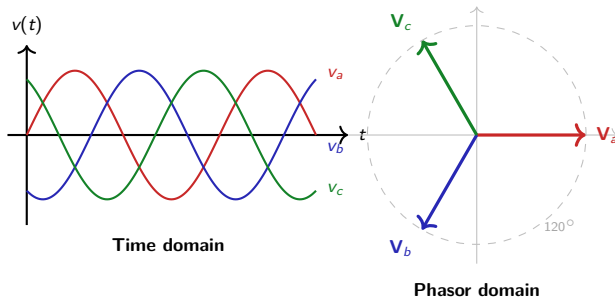
Lecture roadmap

- 1 What “balanced” actually means (with phasor pictures)
- 2 Where symmetry must hold: source, load, network
- 3 The collapse: three coupled equations $\rightarrow$ one
- 4 Single-line diagrams as the visual shortcut
- 5 Positive sequence as the right vocabulary (preview of L04)
- 6 When balanced PF is appropriate
- 7 When it lies – a numerical example

Prerequisite

L01 – Y_{bus} , NR, slack/PV/PQ. L02 helpful but not required.

Balanced three-phase set in pictures



Two equivalent statements of “balanced”

- **Time domain:** three sinusoids of equal amplitude, 120° apart in phase.
- **Phasor domain:** three vectors of equal length, 120° apart in angle.

Both views describe the same physical assumption.

Algebraic statement: the a -operator

Define the rotation operator

$$a = e^{j2\pi/3} = e^{j120^\circ} = -\frac{1}{2} + j\frac{\sqrt{3}}{2}.$$

Then a balanced phase- a , phase- b , phase- c set satisfies

$$\mathbf{V}_b = a^2 \mathbf{V}_a, \quad \mathbf{V}_c = a \mathbf{V}_a$$

And likewise for currents:

$$\mathbf{I}_b = a^2 \mathbf{I}_a, \quad \mathbf{I}_c = a \mathbf{I}_a.$$

Why this matters for computation

If $\mathbf{V}_a$ is known, $\mathbf{V}_b$ and $\mathbf{V}_c$ are determined by multiplication. **One unknown per bus instead of three.** The whole simplification of single-line analysis flows from this identity.

The three places where symmetry must hold

1. Source symmetry

- Generators produce three phases equal in magnitude and 120° apart.
- Transformer windings are symmetric.
- The slack bus is itself balanced.

2. Load symmetry

- Every three-phase load draws equal P , Q on each phase.
- No single-phase laterals.
- No phase-A-only PV.

3. Network symmetry

- Each line has equal self-impedance on all phases.
- Mutual coupling is the same A-B, B-C, A-C.
- Transposed lines or equivalent.

All three or none

The single-line simplification requires *all three* assumptions simultaneously. Break any one and the per-phase quantities drift apart – exactly what L04 will measure.

Where each assumption is typically OK or not

	Source sym.	Load sym.	Network sym.
HV transmission (220+ kV)	✓	✓	✓
MV distribution (10–35 kV)	✓	roughly ✓	roughly ✓
LV three-phase (400 V industrial)	✓	sometimes	sometimes
LV residential (400 V, single-phase laterals)	✓	△	△
LV with single-phase PV / EV / heat pumps	✓	△ △	△
Microgrid (LV)	✓	△ △	△

Reading this table

The further down the table you go, the more single-line analysis becomes a useful first pass rather than a correct answer. By the time you hit the bottom row – which is the regime of this project – you *must* go to L04 to claim phase-specific results.

Three coupled equations collapse to one

Without balance, the three-phase nodal equation at each bus is

$$\begin{bmatrix} \mathbf{I}_a \\ \mathbf{I}_b \\ \mathbf{I}_c \end{bmatrix} = \underbrace{\begin{bmatrix} Y_{aa} & Y_{ab} & Y_{ac} \\ Y_{ba} & Y_{bb} & Y_{bc} \\ Y_{ca} & Y_{cb} & Y_{cc} \end{bmatrix}}_{3 \times 3 \text{ phase-domain admittance}} \begin{bmatrix} \mathbf{V}_a \\ \mathbf{V}_b \\ \mathbf{V}_c \end{bmatrix}$$

For N buses this is a $3N$ -dimensional linear system.

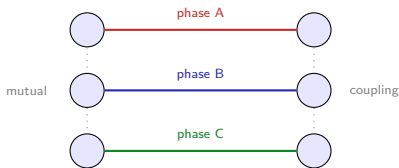
Under the three symmetry assumptions, the 3×3 phase block has the structure $Y_{aa} = Y_{bb} = Y_{cc}$ (self) and $Y_{ab} = Y_{bc} = Y_{ca}$ (mutual). Combined with the rotation identities $\mathbf{V}_b = a^2 \mathbf{V}_a$ and $\mathbf{V}_c = a \mathbf{V}_a$, the three rows above reduce to a *single* equation per bus:

$$\boxed{\mathbf{I}_a = Y^{(1)} \mathbf{V}_a}$$

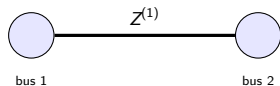
where $Y^{(1)} = Y_{aa} + a^2 Y_{ab} + a Y_{ac}$ is the *positive-sequence admittance* for $V_b = a^2 V_a$, $V_c = a V_a$.

Visual: three-line vs single-line diagram

Three-line diagram



Single-line equivalent



What the collapse buys us

- **Storage:** $3\times$ less state, $9\times$ less admittance matrix.
- **Computation:** a $3N \times 3N$ Jacobian becomes $N \times N$.
- **Cognition:** you can read a one-line diagram on one piece of paper.

The math is L01's math

What you saw in L01:

- Built a single Y_{bus} from line impedances z_{ij} .
- Wrote one PF equation per bus.
- Solved with Newton-Raphson.
- Read `net.res_bus.vm_pu` and `net.res_line.loading_percent`.

What you were actually doing (without knowing the name):

- Building the *positive-sequence* Y_{bus} .
- Solving the *positive-sequence* PF.
- Reading the *positive-sequence* bus voltages.
- Implicitly relying on all three symmetry assumptions.

Why this re-labeling matters

When we get to L04, “positive sequence” is one of three coordinates we will track. The other two – negative and zero sequence – are exactly the modes that single-line analysis throws away.

Three sequence components in one slide

Any three-phase phasor set ($\mathbf{V}_a, \mathbf{V}_b, \mathbf{V}_c$) can be uniquely written as the sum of three balanced sets:

$$\mathbf{V}_a = \mathbf{V}^{(1)} + \mathbf{V}^{(2)} + \mathbf{V}^{(0)}$$

$$\mathbf{V}_b = a^2 \mathbf{V}^{(1)} + a \mathbf{V}^{(2)} + \mathbf{V}^{(0)}$$

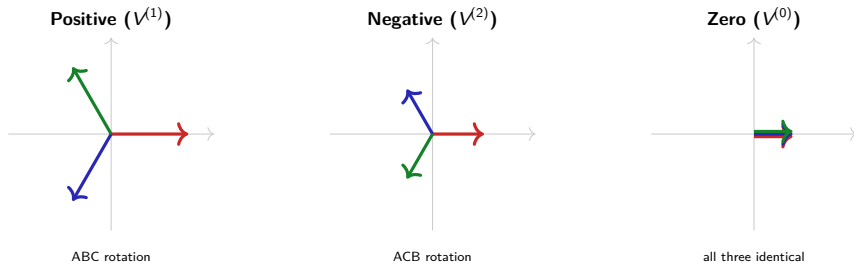
$$\mathbf{V}_c = a \mathbf{V}^{(1)} + a^2 \mathbf{V}^{(2)} + \mathbf{V}^{(0)}$$

Component	Pattern	Meaning
Positive ($V^{(1)}$)	balanced, rotating ABC	true balanced operation
Negative ($V^{(2)}$)	balanced, rotating ACB	a measure of phase asymmetry
Zero ($V^{(0)}$)	all three in phase	a measure of common-mode shift

Key fact for this lecture

Under balanced operation, $\mathbf{V}^{(2)} = \mathbf{V}^{(0)} = 0$. **Balanced PF is positive-sequence PF.** L04 will build all three.

Visualizing the sequence decomposition



Sanity check

Try summing the three positive-sequence phasors at the same instant – they cancel (sum to zero). The same is true for the negative set. The zero-sequence phasors sum to $3V^{(0)}$ – which is why the neutral conductor carries it.

Where balanced PF actually works well

Strong-balance domains:

- **HV transmission** (110–765 kV): transposed lines, three-phase generators, large aggregated loads. Phase-to-phase mismatch typically $< 0.5\%$.
- **Power plants and substations**: three-phase machines, transformers, breakers – inherently symmetric.
- **MV industrial loads**: large motors and arc furnaces, drawing all three phases.
- **Bulk planning studies**: market clearing, OPF, long-term expansion – phase detail is not the question being asked.

Why we still use balanced PF every day

$\sim 3\times$ fewer variables, $\sim 9\times$ smaller admittance matrix, much faster convergence. For most transmission-scale studies, the precision is more than enough.

How to judge whether balance is adequate

Scenario	Expected phase asymmetry	Recommended model
Transposed network with balanced three-phase devices	low	balanced PF is usually appropriate
Planning study using aggregated phase totals	study-dependent	compare against a 3-phase benchmark
LV feeder with single-phase loads, PV, or EV	material	unbalanced 3-phase PF
Neutral/grounding effects or phase-specific protection	material	unbalanced 3-phase PF
Unbalanced faults	intrinsic	sequence/phase-domain fault model

Practical recipe

Do not assume an error percentage from voltage level alone. Compare balanced and three-phase results on representative operating points, then decide whether the approximation preserves the limits relevant to the study.

A diagnostic checklist

If **any** of the following is true, balanced PF can hide real violations:

- ① Loads or DER attached to one or two phases only.
- ② Line parameters that differ phase-to-phase (untransposed LV).
- ③ Neutral conductor effects or grounding asymmetry.
- ④ Fault analysis (faults are inherently unbalanced).
- ⑤ Single-phase protection or per-phase tap controllers.

The microgrid baseline

LV microgrids match items 1, 2, and often 3. Doing balanced PF in this regime and quoting per-phase results would be dishonest. We will not do that in this course.

A numerical example: balanced PF lies

A 2-bus LV feeder. Line: $z = 0.05 + j0.05$ pu (high R/X, LV-like). Slack at $V_1 = 1.00$.

Total load at bus 2 is the same in both scenarios:

Scenario A: balanced load

- 0.133 pu on each of phases A, B, C
- Total $P = 0.4$ pu, balanced

Scenario B: same total, all on phase A

- 0.4 pu on phase A, 0 on B and C
- Total $P = 0.4$ pu, very unbalanced

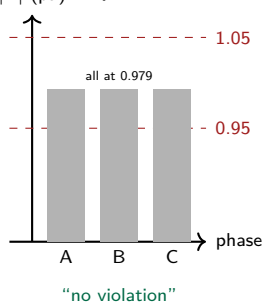
	$ V_a $ at bus 2	$ V_b $ at bus 2	$ V_c $ at bus 2
What balanced PF reports (both scenarios)	0.979	0.979	0.979
What 3 ϕ PF says, Scenario A (truly balanced)	0.979	0.979	0.979
What 3 ϕ PF says, Scenario B (load on A only)	0.908	1.029	0.995

The lie

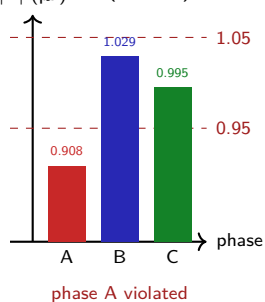
For Scenario B, balanced PF reports “ $V = 0.979$, no violation anywhere”. The true phase-A voltage is 0.908 – well below the 0.95 limit. **Phase A is in violation and the balanced model cannot see it.** L04 will derive these numbers from scratch using symmetrical components.

Visualizing the lie

What balanced PF reports



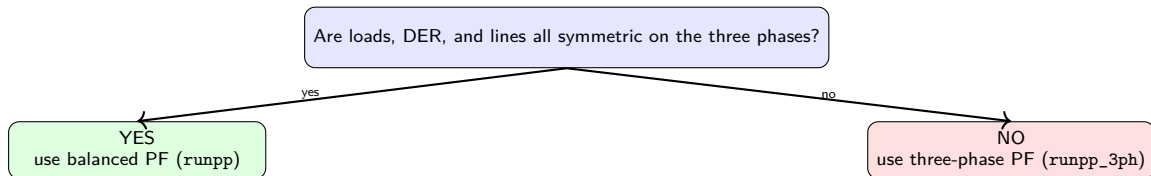
What 3 ϕ PF actually finds (Scen. B)



Reading this picture

Same network. Same total load. Two solvers tell you different stories. In a safety-critical context (which microgrid N-1 screening is), the difference matters: the balanced view would let an operator approve a state that is actually unsafe.

A two-line decision tree



Decision in plain language

If you can answer “yes” confidently and the cost of being wrong is low (planning, market clearing), use balanced PF. If you cannot, or if a hidden violation is consequential (operations, safety screening), pay the extra cost of runpp_3ph.

pandapower preview: same network, two calls

```
import pandapower as pp

# Build a small balanced system (same as L01)
net = pp.create_empty_network(sn_mva=1.0)
b1 = pp.create_bus(net, vn_kv=0.4)
b2 = pp.create_bus(net, vn_kv=0.4)
pp.create_ext_grid(net, bus=b1, vm_pu=1.0,
    s_sc_max_mva=10, rx_max=0.1,
    x0x_max=1.0, r0x0_max=0.1)
pp.create_line_from_parameters(net, b1, b2, length_km=1,
    r_ohm_per_km=0.08, x_ohm_per_km=0.08,
    c_nf_per_km=0.0, max_i_ka=0.4,
    r0_ohm_per_km=0.16, x0_ohm_per_km=0.16,
    c0_nf_per_km=0.0)
pp.create_load(net, bus=b2, p_mw=0.06, q_mvar=0.0)

# --- Balanced PF (single voltage per bus) ---
pp.runpp(net)
print(net.res_bus[["vm_pu", "va_degree"]])

# --- Three-phase PF (one voltage per phase per bus) ---
pp.runpp_3ph(net)
print(net.res_bus_3ph[["vm_a_pu", "vm_b_pu", "vm_c_pu",
    "unbalance_percent"]])
```

What the result columns tell you

res_bus has 1 voltage per bus. res_bus_3ph has 3. Pick the API to match the question you are asking.

Homework

Required

- 1 In pandapower, build a 3-bus balanced LV network. Run `pp.runpp(net)` and `pp.runpp_3ph(net)`. Verify $|V|$ matches across both APIs to 4 decimal places.
- 2 Replace the balanced load with an `asymmetric_load` on phase A only (same total P). Re-run `pp.runpp_3ph`. Report $|V_a|$, $|V_b|$, $|V_c|$.
- 3 Run `pp.runpp` on the same network with the asymmetric load (it will treat it as balanced). Compare to (2). Write 4–6 sentences explaining the discrepancy.
- 4 Read `unbalance_percent` from `res_bus_3ph`. This is the Voltage Unbalance Factor ($VUF = |V^{(2)}|/|V^{(1)}| \times 100\%$, IEC limit 2%; L04 derives this). Is it above 2%?

Optional

- 1 Sweep the phase-A load from 10% to 200% of the per-phase balanced value. Plot $|V_a|$, $|V_b|$, $|V_c|$ on the same axes. At what point does $|V_a|$ cross 0.95?
- 2 Repeat (1) with a delta-connected asymmetric load instead of wye. Explain the difference.

Exit checklist

- ① ✓ I can draw a balanced phasor diagram and state the a -operator identities.
- ② ✓ I can list the three places where symmetry must hold for balanced PF to be valid.
- ③ ✓ I can explain why three coupled phase equations collapse to one under balance.
- ④ ✓ I can translate “balanced PF” into “positive-sequence PF”.
- ⑤ ✓ I can decide between `runpp` and `runpp_3ph` given a description of a network.
- ⑥ ✓ I can construct a numerical example where balanced PF hides a real per-phase violation.
- ⑦ ✓ I am ready for L04 to fill in the sequence-domain math.

References

- **Grainger & Stevenson**, *Power System Analysis*. Chapters on per-phase analysis and symmetrical components.
- **Glover, Sarma, Overbye**, *Power System Analysis and Design*. The per-phase equivalent circuit, plus a clean treatment of why the simplification works.
- **Kersting**, *Distribution System Modeling and Analysis*. Cases where balanced analysis breaks down in distribution.
- **Fortescue, 1918**. *Method of Symmetrical Co-ordinates Applied to the Solution of Polyphase Networks*, AIEE Trans. The original sequence-components paper.
- **pandapower docs**: “Balanced AC Power Flow” and “Asymmetric / Three-phase Power Flow” pages.
- **IEC 61000-4-30**, *Power quality measurement methods*. Negative-to-positive-sequence VUF definition.
- **IEC 61000-2-2**, compatibility levels for low-frequency conducted disturbances in public LV systems.